

Salinity-based information transmission in microfluidic channels

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ABSTRACT

In this paper, we propose a novel, flexible, easy to use and low cost salinity-based micro-scale molecular communication testbed, that allows reliable multi-level information transmission.

CCS CONCEPTS

• **Networks** → **Network experimentation**.

KEYWORDS

Molecular communication, Testbeds, Lab-on-a-Chip

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1 INTRODUCTION

In the last years, many theoretical studies have been published in the field of molecular communications (MC). However, due to the complexity of most MC systems it is inevitable to perform experiments to validate and improve those results. Thus, we propose a novel, flexible, easy to use and low cost micro-scale testbed, which demonstrates salinity-based information transmission on microfluidic chips. In contrast to a similar work in [3], it allows multi-level signaling, can be easily extended to a more complex network of channels and can be used for the precise concentration control in microfluidic channels.

2 PRINCIPLE OF THE TESTBED

2.1 Salinity of Water

The relation between salinity S and conductivity $\sigma(S)$ of water can be approximately described by $\sigma(S) \approx aS$, with

constant a , assuming that the temperature and pressure of the fluid vary insignificantly [2]. If two plain electrodes with conducting surface A are placed at a distance L in the fluid, the conductance of this configuration can be estimated as

$$G(S) = \frac{A\sigma(S)}{L} = \frac{Aa}{L}S = k_0S. \quad (1)$$

Thus, this expression describes the connection between salinity and the measurable electric conductivity.

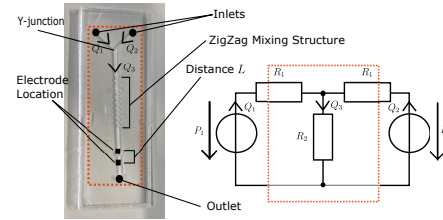


Figure 1: Microfluidic chip and its hydraulic analogy

2.2 Transmitter

The key components for encoding and propagation of salinity-based information are two microfluidic pumps connected to the Y-junction of the microfluidic chip. The microfluidic chip and its hydraulic analogy, with hydraulic resistances R_k , flows Q_i and pressures P_i , is shown in Fig. 1. These pumps can be linearly controlled by altering the electrical voltage applied to them. Thus the pressure P_i created by i th pump can be expressed as $P_i = k_1 V_i$, with the applied voltage V_i and a constant k_1 . Using the hydraulic analogy we can calculate the flows Q_1 and Q_2 as functions of P_1 , P_2 , R_1 and R_2 . The used pumps can only operate with positive input voltage and, thus, the pump pressures are always positive. We demand a constant flow Q_3 in the main channel which ensures, that the corresponding flow velocity is independent of the encoded symbol. To ensure this, the sum of the two pump pressures must remain constant. Consequently, we chose $P_1 = P_0 - \Delta P = k_1(V_0 - \Delta V)$ and $P_2 = P_0 + \Delta P = k_1(V_0 + \Delta V)$, respectively, with $V_0 = 100$ V and $\Delta V \in [0, 14, 28, 42]$ V. Using the hydraulic analogy, Q_1 and Q_2 can be derived by

$$Q_{1,2} = \frac{1}{R_1 + 2R_2} k_1 V_0 \mp \frac{1}{R_1} k_1 \Delta V = A \mp B \Delta V. \quad (2)$$

We assume, that the water transported by Q_1 and Q_2 is pure ($S_1 = 0$) and saltwater ($S_2 > 0$), respectively. For the theoretical derivation of the salinity S_{el} at the electrode location we



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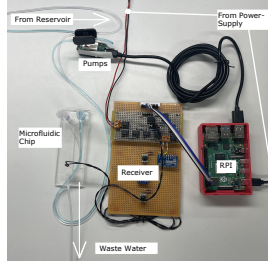


Figure 2: Experimental setup

assume a perfectly working mixing channel. Then, we can use the law of mass-conservation and obtain

$$S_{el} = \frac{Q_1 S_1 + Q_2 S_2}{Q_1 + Q_2} = S_2 \left(0.5 + \frac{B}{2A} \Delta V \right). \quad (3)$$

Based on this relation and on the aforementioned values for ΔV , we can encode 4 symbols in the salinity at the electrodes. It is important to note that the number of symbols can be increased by choosing more values for ΔV .

2.3 Microfluidic Channel

Using numerical values suitable for the proposed setup we calculated the Peclet number and obtained $Pe > 10^4$. Due to its high value the diffusive effects are negligible, which has two implications: *i*) To ensure that (3) is valid we need to include a mixing-structure in the chip design (see [1]); *ii*) For two consecutive blocks of different salinity we can neglect intersymbol interference (ISI) due to diffusive effects. We can estimate the time delay between transmitter and receiver by $t = d/v = d/Q_3/(wh) = \text{const}$, where d denotes the distance between Y-junction and electrode location and w and h correspond to channel width and height, respectively.

2.4 Receiver

In order to measure the salinity S_{el} , given in (3), at the electrodes we designed an AC conductivity measurement circuit. The core of this circuit is an inverting operational amplifier, with a constant resistance R_{fb} in its feedback path and the device under test (i.e., the microfluidic chip) with resistance $R_{el}(S_{el})$ at its input. This circuit is supplied by a sinusoidal input voltage with constant amplitude $\hat{V}_{in}$ and a frequency in the low kHz-range. Thus, the amplitude of the output voltage $\hat{V}_{out}$ depends linearly on the conductivity $G_{el}(S_{el}) = 1/R_{el}(S_{el})$ of the saltwater between the electrodes

$$\hat{V}_{out} = \hat{V}_{in} R_{fb} G_{el}(S_{el}) = k_2 G_{el}(S_{el}). \quad (4)$$

Thus, based on the measured output voltage $\hat{V}_{out}$, we can calculate $G_{el}(S_{el})$ and subsequently the salinity at the electrodes using (1).

3 DEMONSTRATION

We will show the transmission and detection of short text-messages on a microfluidic chip using different salinity levels. Fig. 2 shows the practical realization of the proposed system. The microfluidic pumps are controlled by a RaspberryPi (RPI).

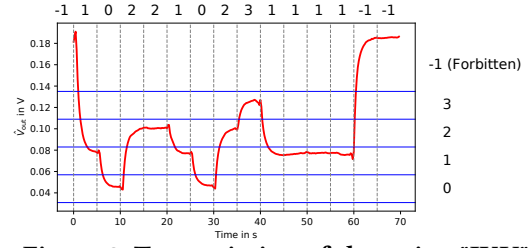


Figure 3: Transmission of the string "JKU"

Depending on the applied voltage they pump pure water and saltwater with different flows from the reservoirs into the chip, resulting in different salinity levels. The electrodes on the chip are connected to the receiver circuit described in Sec. 2.4, which is connected to the RPI over an analog-digital-converter (ADC) for evaluation purposes. The demonstration works as follows: The user enters a string of ASCII characters via a keyboard into a terminal. The RPI maps the character sequence to a bit sequence. In this bit sequence two bits are grouped to form a symbol, resulting in the symbol sequence to be transmitted. We use a forbidden state i.e., a state that is not used during regular transmission to indicate when currently no information is transmitted. The message is sent by adjusting the voltage applied to the microfluidic pumps and is carried across the channel with the constant flow Q_3 . During the transmission, the measured output voltage $\hat{V}_{out}$, the decision thresholds, the detected symbols and the corresponding characters are shown on a monitor. Fig. 3 shows the measured voltages and thresholds (blue lines) for the successful transmission of the string "JKU", which is mapped to the following symbol sequence [1, 0, 2, 2, 1, 0, 2, 3, 1, 1, 1, 1] and corresponds to the ΔV -sequence [14, 0, 28, 28, 14, 0, 28, 42, 14, 14, 14, 14] V. It was recorded using a symbol duration of 5 s and a sampling time of 0.21 s. The conversion factor between the measured output voltage $\hat{V}_{out}$ and the actual salinity S_{el} is given by $1/(k_0 k_2) = 46.7 \text{ mg L}^{-1} \text{ V}^{-1}$. A transient behavior between the symbols can be observed, which is currently the main limiting factor in terms of transmission rate and its cause is currently under investigation.

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